Gihyun Lee

Curriculum Vitae

Department of Chemistry
Purdue University
West Lafayette, IN 47907, United States

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RESEARCH INTERESTS

Research focuses on understanding how stimulus-induced molecular responses are preserved, suppressed, or amplified in crystalline coordination solids. The work combines ligand design, framework synthesis, characterization, and computational approaches to develop stimulus-responsive materials.

ACADEMIC EXPERIENCE

2025 – Present

Purdue University, IN, United States
Postdoctoral Research Associate
Research Advisor: Christopher Uyeda

- Design and synthesis of azoarene-based photoswitchable ligands for incorporation into 2D and 3D coordination frameworks.
- Development of crystallization conditions for coordination materials and investigation of their solid-state photoresponses.
- Perform DFT calculations using Gaussian 16 and GaussView 6 to investigate the conformational energetics of ligands and coordination frameworks through constrained geometry optimization and vibrational frequency analysis.

EDUCATION

2018 – 2025

Yonsei University, South Korea
Ph.D. in Chemistry, February 2025
Research Advisor: Moonhyun Oh
Dissertation: Construction of Hybrid Metal–Organic Frameworks via Structural Adjustment Process

2013 – 2018

University of Victoria, BC, Canada
B.S. in Chemistry, May 2018
Major in Chemistry
Minor in Biochemistry

PUBLICATIONS

*corresponding author   +contributed equally

First- and Co-first author

Gihyun Lee, Dayeon Choi, and Moonhyun Oh*, Activating the Gate-Opening of a Metal–Organic Framework and Maximizing Its Adsorption Capacity. *J. Am. Chem. Soc.* **2025**, 147, 12811−12820. DOI: 10.1021/jacs.5c01399

Hyeongi Lim+, Gihyun Lee+, Jeehyoun Lee, and Moonhyun Oh*, Compositional Complexity of Metal–Organic Frameworks with Programmable Spatial Arrangement of Multi-Metallic Components. *Small* **2025**, 21, 2408119. DOI: 10.1002/smll.202408119 [selected for the back cover]

Sujeong Lee+, Gihyun Lee+, and Moonhyun Oh*, MOF-on-MOF Growth: Inducing Naturally Nonpreferred MOFs and Atypical MOF Growth. *Acc. Chem. Res.* **2024**, *57,* 3113−3125. DOI: 10.1021/acs.accounts.4c00469

Hyunjeong Oh+, Gihyun Lee+, and Moonhyun Oh*, A Drop-and-Drain Method for Convenient and Efficient Fabrication of MOF/Fiber Composites. *Small* **2024**, *20*, 2306543. DOI: 10.1002/smll.202306543

Gihyun Lee+, Sojin Oh+, and Moonhyun Oh*, Enhanced early-stage adsorption of chemical warfare agent simulant by MIL-68-(X%OH). *Bull. Kor. Soc. Chem.* **2024**, *45*, 67−73. DOI: 10.1002/bkcs.12794

Gihyun Lee, Haejin Kwon, Sujeong Lee, and Moonhyun Oh*, Structural Compromise Between Conflicted Spatial-Arrangements of Two Linkers in Metal–Organic Frameworks. *Small Methods*, **2023**, *7*, 2201586. DOI: 10.1002/smtd.202201586 [selected as the cover page]

Sujeong Lee+, Gihyun Lee+, and Moonhyun Oh*, Lattice-Guided Construction and Harvest of a Naturally Nonpreferred Metal−Organic Framework. *ACS Nano*, **2021**, *15*, 17907−17916. DOI: 10.1021/acsnano.1c06207

Jian Yeo+, Gihyun Lee+, Sujeong Lee, and Moonhyun Oh*, Rational manufacture of yolk−shell and core−shell metal oxide double layers from silica-templated coordination polymer double layers. *Mater. Chem. Front.* **2021**, *5*, 3404−3412. DOI: 10.1039/D1QM00034A

Gihyun Lee, Sujeong Lee, Sojin Oh, Dooyoung Kim, and Moonhyun Oh*, Tip-To-Middle Anisotropic MOF-On-MOF Growth with a Structural Adjustment. *J. Am. Chem. Soc.* **2020**, *142*, 3042−3049. DOI: 10.1021/jacs.9b12193

Co-author

Sujeong Lee, Gihyun Lee, and Moonhyun Oh*, Induced Production of Atypical Naturally Non-Preferred Metal–Organic Frameworks and Their Detachment via Provoking Post-Mismatching. *Small*, **2023**, *19*, 2303580. DOI: 10.1002/smll.202303580 [selected for the the inside front cover]

Sojin Oh, Sujeong Lee, Gihyun Lee, and Moonhyun Oh*, Enhanced adsorption capacity of ZIF-8 for chemical warfare agent simulants caused by its morphology and surface charge. *Sci. Rep.* **2023**, *13*, 12250. DOI: 10.1038/s41598-023-39507-6

Sujeong Lee, Sojin Oh, Gihyun Lee, and Moonhyun Oh*, Defective MOF-74 with ancillary open metal sites for the enhanced adsorption of chemical warfare agent simulants. *Dalton Trans.* **2023**, *52*, 12143–12151. DOI: 10.1039/D3DT02025H

Sojin Oh, Sujeong Lee, Gihyun Lee, and Moonhyun Oh*, Boosted ability of ZIF-8 for early-stage adsorption and degradation of chemical warfare agent simulants. *Nanoscale Adv.* **2023**, *5*, 6449–6457. DOI: 10.1039/D3NA00807J

Sunwoo Kwon, Myeong Jin Oh, Soohyun Lee, Gihyun Lee, Insub Jung*, Moonhyun Oh*, and Sungho Park*, Au Octahedral Nanosponges: 3D Plasmonic Nanolenses for Near-Field Focusing. *J. Am. Chem. Soc.* **2023**, *145*, 27397–27406. DOI: 10.1021/jacs.3c08315

Chul Hwan Shim+, Sojin Oh+, Sujeong Lee, Gihyun Lee, and Moonhyun Oh*, Construction of defected MOF-74 with preserved crystallinity for efficient catalytic cyanosilylation of benzaldehyde. *RSC Adv.* **2023**, *13*, 8220–8226. DOI: 10.1039/D3RA01222K

Hyeji Jun+, Sojin Oh+, Gihyun Lee, and Moonhyun Oh*, Enhanced catalytic activity of MOF-74 via providing additional open metal sites for cyanosilylation of aldehydes. *Sci. Rep.* **2022**, *12*, 14735. DOI: 10.1038/s41598-022-18932-z

Elanna B. Stephenson, Ricardo Garcia Ramirez, Sean Farley, Katherine Adolph-Hammond, Gihyun Lee, John M. Frostad, and Katherine S. Elvira*, Investigating the effect of phospholipids on droplet formation and surface property evolution in microfluidic devices for droplet interface bilayer (DIB) formation. *Biomicrofluidics*, **2022**, *16*, 044112. DOI: 10.1063/5.0096193

Dooyoung Kim, Gihyun Lee, Sojin Oh, and Moonhyun Oh*, Unbalanced MOF-on-MOF growth for the production of lopsided core–shell of MIL-88B@MIL-88A with mismatched cell parameters. *Chem. Commun.* **2019**, *55*, 43–46. DOI: 10.1039/C8CC08456D

PATENTS

2023.10.12: Moonhyun Oh, Hyunjeong Oh, and Gihyun Lee. Drop-and-Drain Method for Convenient and Efficient Fabrication of MOF/Fiber Composites. Korean Patent Application # 10-2023-0135860

2026.06.30: Moonhyun Oh, Hyunjeong Oh, and Gihyun Lee. Drop-and-Drain Method for Convenient and Efficient Fabrication of MOF/Fiber Composites. Korean Patent Registration # 10-2986030

SCHOLARSHIPS AND AWARDS

Scholarships

2023.09–2024.12: Teaching Assistant Scholarship (Department of Chemistry, Yonsei University)

2023.09: Graduate Student Idea Incubation Fund: Academic Research Fellowship for "*Small Methods*, **2023**, *7*, 2201586" (Yonsei University)

2018.09–2021.06: Teaching Assistant Scholarship (Department of Chemistry, Yonsei University)

Awards

2026.07: Postdoctoral Supplemental Travel Award (Purdue University, Purdue Postdoctoral Affairs)

2026.01: Outstanding BK21 Participant Recognition Award (Yonsei University, Graduate School)

2022.06: Best Oral Presentation Award for "A Tip-to-Middle Anisotropic MOF-on-MOF Growth with a Structural Adjustment" (Inorganic Chemistry Division of Korean Chemical Society)

2022.04: Excellent Poster Presentation Prize for "Rational Manufacture of Yolk–Shell and Core–Shell Metal Oxide Double Layers from Coordination Polymer Double Layers" (129th General Meeting of the Korean Chemical Society, RSC Poster Prize)

2021.02: Outstanding Teaching Assistant Award (Department of Chemistry, Yonsei University)

2020.06: Merit Academic Paper Award for "*J. Am. Chem. Soc.* **2020**, *142*, 3042–3049" (Yonsei University)

2020.03: Outstanding Teaching Assistant Award (Department of Chemistry, Yonsei University)

2019.08: Outstanding Teaching Assistant Award (Department of Chemistry, Yonsei University)

2016.09: Chemistry Undergraduate Achievement Award: Bronze medal in Chemistry 222 (Department of Chemistry, University of Victoria)

2013.04: Entrance Scholarship (University of Victoria)

TEACHING EXPERIENCE

2023.09–2024.12: Teaching assistant for undergraduates, Department of Chemistry, Yonsei University

2022.03–2023.07: Teaching assistant for a mentorship program in chemistry at the Institute of Science for the Gifted, Yonsei University

2018.09–2021.06: Teaching assistant for undergraduates, Department of Chemistry, Yonsei University

TECHNICAL EXPERTISE

Synthesis and Sample Preparation

Organic ligand synthesis; synthesis of metal–organic frameworks and coordination compounds; air- and moisture-sensitive synthesis using Schlenk-line and glovebox techniques; development of crystallization conditions; single-crystal growth and sample preparation for crystallographic analysis; pellet preparation

Hands-on Instrumentation and Data Analysis

Powder X-ray diffraction (PXRD); scanning electron microscopy with energy-dispersive X-ray spectroscopy (SEM–EDS); optical and fluorescence microscopy; Fourier-transform infrared (FTIR) and Raman spectroscopy; thermogravimetric analysis (TGA); differential scanning calorimetry (DSC); gas adsorption measurements; BET surface-area analysis; NLDFT pore-size distribution analysis; particle and surface zeta-potential measurements; nuclear magnetic resonance (NMR) spectroscopy; UV–Vis and diffuse-reflectance UV–Vis spectroscopy

Additional Data Interpretation Experience

X-ray photoelectron spectroscopy (XPS); transmission electron microscopy (TEM); selected-area electron diffraction (SAED); scanning transmission electron microscopy (STEM); inductively coupled plasma analysis (ICP); mass spectrometry (MS)

Computational Chemistry

Gaussian 16 and GaussView 6; constrained DFT geometry optimization; vibrational frequency analysis; molecular-structure visualization and comparative energy analysis

LIST OF REFEREES

Contact information for referees will be provided upon request.